Syllabus Financial Econometrics
MSc Finance Suite Imperial Business School

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1 Course Aims

This module will familiarize students with a range of modern econometric techniques essential to empirical research in finance. At the end of the module, students will be able to:

1. construct adequate econometric models;
2. estimate the chosen econometric models;
3. evaluate the outputs of those models;
4. use the models to inform financial decisions.

2 Course Structure

The course will last 8 weeks plus one week revision. Each week will have two 1.5 hour lectures and a class. Lecture notes and problem sets will be posted on Isendi each week. Students will have to solve the problems before classes and then discuss them in the class with the TA. It is important that students make an effort to read the core reading before the lectures, so that they have a first understanding of the topics at hand.

3 Course Material

The main textbook is

- Brooks, C. (2019). *Introductory Econometrics for Finance*. Fourth edition. Cambridge University Press, Cambridge. (Henceforth Brooks, or B)

We will also read from

- James, G., Witten, D., Hastie, T. and Tibshirani, R. *Introduction to Statistical Learning with Applications in R*. 2nd Ed. (Henceforth ISLR)

ISLR can be found for free on www.statlearning.com, together with the companion book *An Introduction to Statistical Learning with Applications in Python*.

For those who have had very little statistics in their undergraduate degree, the following textbook provides a primer on the topic.

- Gujarati and Porter (2009). *Basic Econometrics*. Fifth edition. McGraw-Hill.

In addition, supplementary reading material will be given during each lecture.

4 Assessment

The assessment is split into two parts

1. 75% Final Exam: Closed book with open questions. A mock exam will be posted on Isendi.
2. 20% Group Coursework: Using real world databases and applying methodologies taught in class.
3. 5% Group Presentation: Presenting your work in a 5 minute presentation.

5 Course Outline

Below is a provisional course outline with compulsory reading.

Week	Topic	Core Reading
1	Linear Regression	B3
2	Multiple Regression	B4, B5
3	Model Selection	ISLR5, ISLR6
4	Univariate TS (ARMA)	B6
5	Unit Roots	B8
6	Cointegration	B8
7	Volatility	B9
8	Presentations	-
9	Revision	-

B = Brooks, *Financial Econometrics*
ISLR = *Introduction to Statistical Learning*

Week 1: Regression

This week we start with the Classic Linear Regression Model (CLRM). Also referred to as **Ordinary Least Squares (OLS)**, it remains the cornerstone of econometrics. Key statistical concepts like hypothesis testing, confidence intervals, and p-values are introduced in the context of regression analysis. We will discuss desirable statistical properties, including unbiasedness, consistency, and efficiency and key assumptions, many covered by the Gauss-Markov theorem.

Required Reading

- Brooks, Chapter 3

Supplementary Reading

- Brooks, Chapter 1 and 2
- ISLR, Chapter 3
- Jensen, M. (1968). The performance of mutual funds in the period 1945-1964. *Journal of Finance*, **23**, 389-416.

Week 2 and 3: Multiple Regression

This week we allow for the inclusion of more than one explanatory variable. We cover model estimation using the OLS method in a **multivariate** context, the interpretation of regression coefficients, and hypothesis testing through the F-test. The session also discusses key diagnostic measures such as R^2 and adjusted R^2, illustrating their usefulness and limitations. We will also look into whether our econometric models are valid by conducting **diagnostic tests**. We discuss the potential issues that can arise when the assumptions of the Classical Linear Regression Model (CLRM) are violated. We learn how to detect problems like heteroskedasticity, auto-correlation, multi-collinearity, and misspecified functional forms.

Required Reading

- Brooks, Chapter 4
- Brooks, Chapter 5

Supplementary Reading

- Hasanhodzic, J. and A. Lo (2007). Can Hedge-Fund Returns be Replicated?: The Linear Case. *Journal of Investment Management*, **5**, 5-45.

Week 4: Model Selection

This week explores the challenges of **model selection and overfitting** in financial econometrics, drawing on concepts from An Introduction to Statistical Learning (Chs. 5–6). We begin by asking how to decide which predictors to include in a regression, contrasting traditional approaches such as adjusted R^2, F-tests, and information criteria (AIC, BIC) with modern out of sample techniques like cross-validation. We then turn to the bias–variance trade-off and illustrate how overly complex models can fit noise rather than signal, a pervasive risk in empirical finance where datasets are high-dimensional and noisy. Shrinkage methods such as ridge regression and lasso are introduced as tools that improve out-of-sample performance by penalising complexity, helping us balance interpretability and predictive accuracy. The week closes with financial applications, showing how different selection and regularisation strategies affect the identification of relevant risk factors and the robustness of asset pricing models.

Required Reading

- ISLR, Chapter 5
- ISLR, Chapter 6

Supplementary Reading

- Cantor, R. and F. Packer (1996). Determinants and impact of sovereign credit ratings. *Federal Reserve Bank of New York Research Paper*.

Week 5: Univariate Time Series

This week, we start our journey into time series econometrics. We will explore how to model and forecast financial time series using **Autoregressive Moving Average (ARMA) models**. We learn how to identify key time series features such as trends, and autocorrelation. Using the Box-Jenkins methodology, we walk through the process of model selection, estimation, and diagnostic checking. We also discuss how to forecast future values using ARMA models and evaluate model performance using forecasting accuracy measures like MSE and MAE. Along the way, we emphasise the balance between statistical models and real-world forecasting challenges, including the limitations imposed by data noise and structural changes.

Required Reading

- Brooks, Chapter 6

Supplementary Reading

- Gerlow, M.E., Irwin, S.H. and T.-R. Liu (1993). Economic Evaluation of Commodity Price Forecasting Models. *International Journal of Forecasting*, 9, 387-397.
- Diebold, F.X. (2007). *Elements of Forecasting*. Thomson, Mason.

Week 6: Unit Roots

This week, we focus on unit roots and **non-stationarity** in time series data, crucial concepts for understanding long-term economic and financial trends. We learn how non-stationary processes, such as random walks, can lead to spurious regressions if not properly addressed. We explore methods for detecting **unit roots** using tests like the Augmented Dickey-Fuller (ADF), Phillips-Perron (PP), and KPSS tests, along with strategies for dealing with structural breaks. We also consider cases where non-stationarity arises due to seasonal patterns, highlighting the need for careful model specification in time series econometrics.

Required Reading

- Brooks, Chapter 8

Supplementary Reading

- Gatev, E., W. N. Goetzmann and K. G. Rouwenhorst (2006). Pairs Trading: Performance of a Relative-Value Arbitrage Rule. *Review of Financial Studies*, **19**, 797 - 827.

Week 7: Cointegration

This week, we explore the concept of **cointegration**, which helps identify long-run equilibrium relationships between non-stationary time series. We discuss how economic and financial variables, such as stock prices and dividends, can move together even if individually non-stationary. We introduce the Engle-Granger two-step method and its extensions, as well as the Johansen procedure, which allows testing for multiple cointegrating vectors in a multivariate setting. Practical examples from finance, such as spot and futures prices, highlight how cointegration supports market efficiency and the absence of arbitrage opportunities.

Required Reading

- Brooks, Chapter 8

Supplementary Reading

- Gatev, E., W. N. Goetzmann and K. G. Rouwenhorst (2006). Pairs Trading: Performance of a Relative-Value Arbitrage Rule. *Review of Financial Studies*, **19**, 797 - 827.

Week 8: Volatility

This week, we focus on modeling financial volatility using **Generalised Autoregressive Conditional Heteroskedasticity (GARCH)** models. We explore how financial returns exhibit key characteristics such as volatility clustering, leverage effects, and fat tails, which standard OLS cannot handle due to heteroskedasticity. We cover the development from ARCH models to GARCH, including extensions like GJR-GARCH and EGARCH, which account for asymmetric shocks. Practical applications include risk management, option pricing, and forecasting, demonstrating how GARCH-based models inform financial decision-making.

During the tutorials of this week, we will have student presentations; where students present the results of their econometrics projects.

Required Reading

- Brooks, Chapter 9

Supplementary Reading

- Day, T. and C. Lewis (1992). Stock Market Volatility and the Information Content of Stock Index Options. Journal of Econometrics, 52, 267-287
- Brooks, C. Henry, O. and G. Persand (2002). The Effect of Asymmetries on Optimal Hedge Ratios. The Journal of Business, 75(2), 333-352.

Week 9: Revision

This week is dedicated to revision and preparing for your upcoming exam. We will review key concepts from the module and discuss an old exam to help you familiarize yourself with the format and the types of questions you can expect. This session will give you the opportunity to clarify any lingering doubts and strengthen your understanding of the material.

In addition, we encourage you to actively participate in the Eddiscussion platform, where you can post any exam-related questions. This will be a valuable resource for you to engage with your peers and get answers to any queries you might have leading up to the exam.